

THE UNIFORM $\mathbb{G}_0$ DICHOTOMY

B. MILLER

ABSTRACT. We give a classical proof of a generalization of the $\mathbb{G}_0$ dichotomy to uniformly analytic families of digraphs indexed by elements of $\kappa^{\mathbb{N}}$ and obtain Kanovei's generalization of the $\mathbb{G}_0$ dichotomy to κ -Souslin digraphs as a corollary.

INTRODUCTION

Suppose that κ is an infinite cardinal and X is a Hausdorff space. Endow κ with the discrete topology and $\kappa^{\mathbb{N}}$ with the product topology. For all $n \in \mathbb{N}$ and $s \in \kappa^n$, set $|s| = n$. For all $\sigma \in \kappa^{\leq \mathbb{N}}$, let $s \sqsubseteq \sigma$ indicate that $\forall i < |s| \ s(i) = \sigma(i)$. Define $\mathcal{N}_s = \{\sigma \in \kappa^{\mathbb{N}} \mid s \sqsubseteq \sigma\}$. We write $\forall^* \sigma \in \kappa^{\mathbb{N}} \ \phi(\sigma)$ to indicate that $\{\sigma \in \kappa^{\mathbb{N}} \mid \phi(\sigma)\}$ is comeager. The *projection* function from $\prod_{i \in I} X_i$ to X_i is given by $\text{proj}_{X_i}(x) = x(i)$. The *horizontal and vertical sections* of a set $R \subseteq X \times Y$ are given by $R^y = \{x \in X \mid x R y\}$ and $R_x = \{y \in Y \mid x R y\}$. Given a function $\pi: R \subseteq X \times Y \rightarrow Z$, define $\pi_x: R_x \rightarrow Z$ by $\pi_x(y) = \pi(x, y)$.

We say that $\phi: C \subseteq \kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}} \rightarrow \kappa^{\mathbb{N}} \times X$ is *parameter preserving* if $\text{proj}_{\kappa^{\mathbb{N}}} \circ \phi = \text{proj}_{\kappa^{\mathbb{N}}} \upharpoonright C$, and $A \subseteq \kappa^{\mathbb{N}} \times X$ has *uniformly analytic (vertical) sections* if there are a closed set $C \subseteq \kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ and a continuous parameter-preserving function $\phi: C \rightarrow \kappa^{\mathbb{N}} \times X$ such that $A = \phi(C)$.

The *diagonal union* of a sequence $(X_s)_{s \in \kappa^{< \mathbb{N}}}$ of subsets of $\kappa^{\mathbb{N}} \times X$ is given by $\nabla_{s \in \kappa^{< \mathbb{N}}} X_s = \bigcup_{s \in \kappa^{< \mathbb{N}}} X_s \cap (\mathcal{N}_s \times X)$. Countable unions are expressible as diagonal unions, and the vertical sections of diagonal unions are themselves countable unions. We say that a set $B \subseteq \kappa^{\mathbb{N}} \times X$ has *uniformly Borel (vertical) sections* if it is in the smallest family of subsets of $\kappa^{\mathbb{N}} \times X$ containing the products of $\kappa^{\mathbb{N}}$ with open subsets of X and closed under complements and diagonal unions. We say that a function $c: \kappa^{\mathbb{N}} \times X \rightarrow \mathbb{N}$ is *uniformly Borel* if $c^{-1}(\{n\})$ has uniformly Borel sections for all $n \in \mathbb{N}$.

A digraph on X is an irreflexive set $G \subseteq X \times X$. A *homomorphism* from a digraph G on X to a digraph H on Y is a function $\pi: X \rightarrow Y$ such that $(\pi \times \pi)(G) \subseteq H$. A *Y -coloring* of G is a homomorphism $c: X \rightarrow Y$ from G to inequality. Fix $\mathfrak{s} \in \prod_{n \in \mathbb{N}} 2^n$ such that $\mathfrak{s}(\mathbb{N})$ is dense in $2^{< \mathbb{N}}$ and define $\mathbb{G}_0 = \bigcup_{n \in \mathbb{N}} \{(\mathfrak{s}_n \frown (i) \frown c)_{i < 2} \mid c \in 2^{\mathbb{N}}\}$.

Here we establish the following version of the $\mathbb{G}_0$ dichotomy:

Theorem 1. *Suppose that κ is an infinite cardinal, X is a Hausdorff space, and $G \subseteq \kappa^{\mathbb{N}} \times (X \times X)$ has uniformly analytic sections that are digraphs on X . Then exactly one of the following holds:*

- (1) *There is a uniformly Borel function $c: \kappa^{\mathbb{N}} \times X \rightarrow \mathbb{N}$ such that c_σ is a coloring of G_σ for all $\sigma \in \kappa^{\mathbb{N}}$.*
- (2) *There is a continuous homomorphism $\pi: 2^{\mathbb{N}} \rightarrow X$ from $\mathbb{G}_0$ to G_σ for some $\sigma \in \kappa^{\mathbb{N}}$.*

In §1, we note the basic closure properties of sets with uniformly analytic and Borel sections. In §2, we consider how these notions interact with Baire category. In §3, we establish a uniform version of Lusin's separation theorem. In §4, we prove Theorem 1. And in §5, we obtain Kanovei's κ -Souslin generalization [Kan97] of the $\mathbb{G}_0$ dichotomy.

We leave it to the reader to check that these results can be combined with the arguments establishing the usual consequences of the $\mathbb{G}_0$ dichotomy to obtain analogous uniform and κ -Souslin generalizations.

Under $\text{ZF} + \text{DC}$, the arguments can be pushed through by retaining codes for the relevant objects and replacing the choices of representations and separating sets by canonical operations on these codes. We avoid this approach, however, in favor of a clearer presentation.

Under $\text{ZF} + \text{DC} + \text{Det}(\Delta_{2n}^1)$, a more careful version of this book-keeping likely yields the generalization of the $\mathbb{G}_0$ dichotomy to Σ_{2n+1}^1 digraphs. The rough idea would be to use higher sharps and the associated companion theorems to obtain canonical tree representations and parameter-dependent admissible sets whose Σ_1 -definable subsets of $\mathbb{N}^{\mathbb{N}}$ have the required complexity. Carrying out the coded version of the arguments inside these admissible sets should then yield Δ_{2n+1}^1 prewellorderings whose equivalence classes are G -independent.

1. POINTCLASSES

We direct the reader to [Kec95, §14] for the most basic facts concerning analytic sets. The following closure properties can be proven in the same fashion as the corresponding facts for analytic sets:

Proposition 1.1. *Suppose that κ is an aleph and X is a Hausdorff space. Then the subsets of $\kappa^{\mathbb{N}} \times X$ with uniformly analytic sections are closed under countable intersections and countable unions.*

Proposition 1.2. *Suppose that κ is an aleph, X and Y are Hausdorff spaces, $A \subseteq \kappa^{\mathbb{N}} \times X$ and $B \subseteq \kappa^{\mathbb{N}} \times Y$ have uniformly analytic sections, and $f: A \rightarrow \kappa^{\mathbb{N}} \times Y$ is continuous and parameter preserving. Then $f(A)$ and $f^{-1}(B)$ have uniformly analytic sections.*

Given a set $S \subseteq \kappa^{<\mathbb{N}}$ and a sequence $(X_s)_{s \in S}$ of subsets of $\kappa^{\mathbb{N}} \times X$, define $\nabla_{s \in S} X_s = \nabla_{s \in \kappa^{<\mathbb{N}}} X_s$, where $X_s = \emptyset$ for all $s \in \sim S$.

Proposition 1.3. *Suppose that κ is an aleph and X is a Hausdorff space. Then the subsets of $\kappa^{\mathbb{N}} \times X$ with uniformly analytic sections are closed under diagonal unions.*

Proof. Suppose that $(A_s)_{s \in \kappa^{<\mathbb{N}}}$ is a sequence of subsets of $\kappa^{\mathbb{N}} \times X$ with uniformly analytic sections and $A = \nabla_{s \in \kappa^{<\mathbb{N}}} A_s$. Then there are closed sets $C_s \subseteq \kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ and continuous parameter-preserving functions $\phi_s: C_s \rightarrow \kappa^{\mathbb{N}} \times X$ with $A_s = \phi_s(C_s)$ for all $s \in \kappa^{<\mathbb{N}}$. As the set $C_n = \nabla_{s \in \kappa^n} C_s$ is closed and the function $\phi_n = \bigcup_{s \in \kappa^n} \phi_s \upharpoonright (C_s \cap (\mathcal{N}_s \times \mathbb{N}^{\mathbb{N}}))$ is continuous, the set $A_n = \nabla_{s \in \kappa^n} A_s = \phi_n(C_n)$ has uniformly analytic sections for all $n \in \mathbb{N}$, as does $A = \bigcup_{n \in \mathbb{N}} A_n$ by Proposition 1.1. $\square$

Proposition 1.4. *Suppose that κ is an aleph and X is a topological space. Then every Borel subset of $\kappa^{\mathbb{N}} \times X$ has uniformly Borel sections.*

Proof. It is sufficient to show that every open set $U \subseteq \kappa^{\mathbb{N}} \times X$ has uniformly Borel sections. For all $s \in \kappa^{<\mathbb{N}}$, let $U_s \subseteq X$ be the maximal open set for which $\mathcal{N}_s \times U_s \subseteq U$. Then $U = \nabla_{s \in \kappa^{<\mathbb{N}}} \kappa^{\mathbb{N}} \times U_s$. $\square$

Proposition 1.5. *Suppose that κ is an aleph, X and Y are topological spaces, $B \subseteq \kappa^{\mathbb{N}} \times Y$ has uniformly Borel sections, and $f: \kappa^{\mathbb{N}} \times X \rightarrow \kappa^{\mathbb{N}} \times Y$ is Borel and parameter preserving. Then $f^{-1}(B)$ has uniformly Borel sections.*

Proof. The family of sets for which the proposition holds contains the open sets by Proposition 1.4 and is clearly closed under complements. The fact that f is parameter preserving ensures that it is also closed under diagonal unions. $\square$

Proposition 1.6. *Suppose that X is a Hausdorff space, $A \subseteq \kappa^{\mathbb{N}} \times X$ has uniformly analytic sections, and $B \subseteq \kappa^{\mathbb{N}} \times X$ has uniformly Borel sections. Then $A \cap B$ and $A \setminus B$ have uniformly analytic sections.*

Proof. By the usual proof of the corresponding fact for analytic and Borel sets, we need only note that if $B = \nabla_{s \in \kappa^{<\mathbb{N}}} B_s$ and both $A \cap B_s$ and $A \setminus B_s$ have uniformly analytic sections for all $s \in \kappa^{<\mathbb{N}}$, then $A \cap B = \nabla_{s \in \kappa^{<\mathbb{N}}} A \cap B_s$ has uniformly analytic sections by Proposition 1.3, and $A \setminus B = \bigcap_{n \in \mathbb{N}} \nabla_{s \in \kappa^n} A \setminus B_s$ has uniformly analytic sections by Propositions 1.1 and 1.3. $\square$

2. BAIRE CATEGORY

We direct the reader to [Kec95, §8] for the most basic facts concerning Baire category (such as localization). Although $\kappa^{\mathbb{N}}$ need not be

separable, the usual proofs use only the Baire category theorem and go through unchanged for completely metrizable spaces.

Proposition 2.1. *Suppose that κ is an aleph, X is a topological space, and $B \subseteq \kappa^{\mathbb{N}} \times X$ has uniformly Borel sections. Then every horizontal section of B has the Baire property.*

Proof. It is enough to show that if $n \in \mathbb{N}$ and the horizontal sections of $B_s \subseteq \kappa^{\mathbb{N}} \times X$ have the Baire property for all $s \in \kappa^{<\mathbb{N}}$, then the horizontal sections of $B = \bigcap_{s \in \kappa^n} B_s$ also have the Baire property. Given $x \in X$, fix an open set $U_s \subseteq \kappa^{\mathbb{N}}$ and dense open sets $U_{m,s} \subseteq \kappa^{\mathbb{N}}$ such that $(B_s^x \triangle U_s) \cap \bigcap_{m \in \mathbb{N}} U_{m,s} = \emptyset$ for all $s \in \kappa^n$. Then $U = \bigcup_{s \in \kappa^n} \mathcal{N}_s \cap U_s$ is open, $U_m = \bigcup_{s \in \kappa^n} \mathcal{N}_s \cap U_{m,s}$ is dense and open for all $m \in \mathbb{N}$, and $(B^x \triangle U) \cap \bigcap_{m \in \mathbb{N}} U_m = \emptyset$, so B^x has the Baire property. $\square$

3. SEPARATION

Recall that $B \subseteq X$ separates $A_0, A_1 \subseteq X$ if $A_0 \subseteq B$ and $A_1 \cap B = \emptyset$.

Proposition 3.1. *Suppose that κ is an aleph, X is a Hausdorff space, and $A_0, A_1 \subseteq \kappa^{\mathbb{N}} \times X$ are disjoint sets with uniformly analytic sections. Then they are separated by a set with uniformly Borel sections.*

Proof. Fix closed sets $C_i \subseteq \kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ and continuous parameter-preserving functions $\phi_i: C_i \rightarrow \kappa^{\mathbb{N}} \times X$ such that $A_i = \phi_i(C_i)$, and define $A_{i,s,t} = \phi_i(C_i \cap (\mathcal{N}_s \times \mathcal{N}_t))$ for all $i < 2$, $s \in \kappa^{<\mathbb{N}}$, and $t \in \mathbb{N}^{<\mathbb{N}}$.

Following the usual convention, we identify $\bigcup_{n \in \mathbb{N}} \kappa^n \times \mathbb{N}^n \times \mathbb{N}^n$ with $(\kappa \times \mathbb{N} \times \mathbb{N})^{<\mathbb{N}}$. Set $T = \{(s, t, u) \in (\kappa \times \mathbb{N} \times \mathbb{N})^{<\mathbb{N}} \mid A_{0,s,t} \cap \overline{A_{1,s,u}} \neq \emptyset\}$.

Lemma 3.2. *The tree T is well-founded.*

Proof. Suppose, towards a contradiction, that $(\sigma, c_0, c_1) \in [T]$. Then $(\sigma, c_i) \in C_i$ for all $i < 2$. Set $x_i = (\text{proj}_X \circ \phi_i)(\sigma, c_i)$ for all $i < 2$. As A_0 and A_1 are disjoint, it follows that $x_0 \neq x_1$, so there are disjoint open neighborhoods U_0 and U_1 of x_0 and x_1 . Fix $n \in \mathbb{N}$ such that $A_{i,\sigma \upharpoonright n, c_i \upharpoonright n} \subseteq \kappa^{\mathbb{N}} \times U_i$ for all $i < 2$. Then $A_{0,\sigma \upharpoonright n, c_0 \upharpoonright n} \cap \overline{A_{1,\sigma \upharpoonright n, c_1 \upharpoonright n}} = \emptyset$, which contradicts the fact that $(\sigma \upharpoonright n, c_0 \upharpoonright n, c_1 \upharpoonright n) \in T$. $\square$

Recursively define $B_{s,t,u} \subseteq \kappa^{\mathbb{N}} \times X$ for $(s, t, u) \in (\kappa \times \mathbb{N} \times \mathbb{N})^{<\mathbb{N}}$ by

$$B_{s,t,u} = \begin{cases} \bigcap_{r \in S_s} \bigcup_{m \in \mathbb{N}} \bigcap_{n \in \mathbb{N}} B_{r, t \hat{\smallfrown} (m), u \hat{\smallfrown} (n)} & \text{if } (s, t, u) \in T \text{ and} \\ \sim \overline{A_{1,s,u}} & \text{otherwise,} \end{cases}$$

where $S_s = \{s \hat{\smallfrown} (\alpha) \mid \alpha < \kappa\}$. The obvious induction shows that $B_{s,t,u}$ has uniformly Borel sections and separates $A_{0,s,t}$ from $A_{1,s,u}$, so $B_{\emptyset, \emptyset, \emptyset}$ separates A_0 from A_1 . $\square$

Given sets $A_i \subseteq \kappa^{\mathbb{N}} \times X_i$ for all $i < 2$ and $R \subseteq \kappa^{\mathbb{N}} \times (X_0 \times X_1)$, we say that (A_0, A_1) is *R-independent* if $\forall \sigma \in \kappa^{\mathbb{N}} ((A_0)_\sigma \times (A_1)_\sigma) \cap R_\sigma = \emptyset$.

Proposition 3.3. *Suppose that κ is an aleph, X_0, X_1 are Hausdorff spaces, $A_i \subseteq \kappa^{\mathbb{N}} \times X_i$ for all $i < 2$ and $R \subseteq \kappa^{\mathbb{N}} \times (X_0 \times X_1)$ have uniformly analytic sections, and (A_0, A_1) is R-independent. Then there are sets $B_i \subseteq \kappa^{\mathbb{N}} \times X_i$ for all $i < 2$ with uniformly Borel sections such that $A_0 \subseteq B_0$, $A_1 \subseteq B_1$, and (B_0, B_1) is R-independent.*

Proof. The set $A'_0 = \text{proj}_{\kappa^{\mathbb{N}} \times X_0}((\text{proj}_{\kappa^{\mathbb{N}} \times X_1} \upharpoonright R)^{-1}(A_1))$ has uniformly analytic sections by two applications of Proposition 1.2. As $A_0 \cap A'_0 = \emptyset$, Proposition 3.1 yields a set $B_0 \subseteq \kappa^{\mathbb{N}} \times X_0$ with uniformly Borel sections separating A_0 from A'_0 .

Along similar lines, the set $A'_1 = \text{proj}_{\kappa^{\mathbb{N}} \times X_1}((\text{proj}_{\kappa^{\mathbb{N}} \times X_0} \upharpoonright R)^{-1}(B_0))$ has uniformly analytic sections by Propositions 1.2, 1.5, and 1.6. As $A_1 \cap A'_1 = \emptyset$, Proposition 3.1 yields a set $B_1 \subseteq \kappa^{\mathbb{N}} \times X_1$ with uniformly Borel sections separating A_1 from A'_1 . $\square$

Given a set $A \subseteq \kappa^{\mathbb{N}} \times X$ and $G \subseteq \kappa^{\mathbb{N}} \times (X \times X)$, we say that A is *G-independent* if (A, A) is G-independent.

Proposition 3.4. *Suppose that κ is an aleph, X is a Hausdorff space, $A \subseteq \kappa^{\mathbb{N}} \times X$ and $G \subseteq \kappa^{\mathbb{N}} \times (X \times X)$ have uniformly analytic sections, and A is G-independent. Then there exists $B \subseteq \kappa^{\mathbb{N}} \times X$ with uniformly Borel sections such that $A \subseteq B$ and B is G-independent.*

Proof. By Proposition 3.3, there exist $B_0, B_1 \subseteq \kappa^{\mathbb{N}} \times X$ with uniformly Borel sections such that $A \subseteq B_0 \cap B_1$ and (B_0, B_1) is G-independent. Then the set $B_0 \cap B_1$ is as desired. $\square$

4. THE UNIFORM $\mathbb{G}_0$ DICHOTOMY

We use the notation and basic facts concerning $\mathbb{G}_0$, colorings, and homomorphisms from [dRM21, §1].

Proof of Theorem 1. Define $p_i: \kappa^{\mathbb{N}} \times (X \times X) \rightarrow \kappa^{\mathbb{N}} \times X$ by setting $p_i(\sigma, (x_0, x_1)) = (\sigma, x_i)$ for all $i < 2$, $\sigma \in \kappa^{\mathbb{N}}$, and $x_0, x_1 \in X$. As Propositions 1.1 and 1.2 ensure that $\bigcup_{i < 2} p_i(G)$ has uniformly analytic sections, there are closed sets $C, D \subseteq \kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ and continuous parameter-preserving functions $\phi: C \rightarrow \kappa^{\mathbb{N}} \times X$ and $\psi: D \rightarrow \kappa^{\mathbb{N}} \times (X \times X)$ such that $\bigcup_{i < 2} p_i(G) = \phi(C)$ and $G = \psi(D)$.

An *approximation* is a triple $a = (n^a, \phi^a, (\psi_k^a)_{k < n^a})$, where $n^a \in \mathbb{N}$, $\phi^a: 2^{n^a} \rightarrow \mathbb{N}^{n^a}$, and $\psi_k^a: 2^{n^a - (k+1)} \rightarrow \mathbb{N}^{n^a}$ for all $k < n^a$. An approximation b is a *one-step extension* of a if:

- (1) $n^b = n^a + 1$,

- (2) $\forall u \in 2^{n^a} \forall v \in 2^{n^b} (u \sqsubseteq v \implies \phi^a(u) \sqsubseteq \phi^b(v))$, and
- (3) $\forall k < n^a \forall u \in 2^{n^a - (k+1)} \forall v \in 2^{n^b - (k+1)} (u \sqsubseteq v \implies \psi_k^a(u) \sqsubseteq \psi_k^b(v))$.

For all $\sigma \in \kappa^{\mathbb{N}}$, a σ -configuration is a triple $\gamma = (n^\gamma, \phi^\gamma, (\psi_k^\gamma)_{k < n^\gamma})$, where $\phi^\gamma: 2^{n^\gamma} \rightarrow \mathbb{N}^{\mathbb{N}}$ and $\psi_k^\gamma: 2^{n^\gamma - (k+1)} \rightarrow \mathbb{N}^{\mathbb{N}}$ for all $k < n^\gamma$, such that:

- (1) $\forall u \in 2^{n^\gamma} \phi^\gamma(u) \in C_\sigma$,
- (2) $\forall k < n^\gamma \forall v \in 2^{n^\gamma - (k+1)} \psi_k^\gamma(v) \in D_\sigma$, and
- (3) $\forall k < n^\gamma \forall v \in 2^{n^\gamma - (k+1)}$
 $(\text{proj}_{X \times X} \circ \psi_\sigma \circ \psi_k^\gamma)(v) = ((\text{proj}_X \circ \phi_\sigma \circ \phi^\gamma)(s_k \frown (i) \frown v))_{i < 2}$.

We say that γ is *compatible* with $Y \subseteq X$ if $(\text{proj}_X \circ \phi_\sigma \circ \phi^\gamma)(2^{n^\gamma}) \subseteq Y$, and *compatible* with a if:

- (1) $n^\gamma = n^a$,
- (2) $\forall u \in 2^{n^a} \phi^a(u) \sqsubseteq \phi^\gamma(u)$, and
- (3) $\forall k < n^a \forall v \in 2^{n^a - (k+1)} \psi_k^a(v) \sqsubseteq \psi_k^\gamma(v)$.

Given a set $Y \subseteq \kappa^{\mathbb{N}} \times X$, we say that a is Y_σ -terminal if no σ -configuration is compatible with Y_σ and a one-step extension of a . Let $A(a, Y)$ denote the set of pairs of the form $(\sigma, (\phi_\sigma \circ \phi^\gamma)(s_{n^a}))$, where $\sigma \in \kappa^{\mathbb{N}}$ and γ is a σ -configuration compatible with a and Y_σ .

The following fact can be proven in the exact same way as in the classical proof of the $\mathbb{G}_0$ dichotomy (see [dRM21, Lemma 1.5]):

Lemma 4.1. *Suppose that a is an approximation, $\sigma \in \kappa^{\mathbb{N}}$, $Y \subseteq \kappa^{\mathbb{N}} \times X$, and a is Y_σ -terminal. Then $A(a, Y)_\sigma$ is G_σ -independent.*

Lemma 4.2. *If a is an approximation and $B \subseteq \kappa^{\mathbb{N}} \times X$ has uniformly Borel sections, then $A(a, B)$ has uniformly analytic sections.*

Proof. The set of pairs (σ, γ) for which γ is a σ -configuration compatible with a is closed, and therefore has uniformly Borel sections by Proposition 1.4, so the set of such pairs for which γ is also compatible with B_σ has uniformly Borel sections by Proposition 1.5, and therefore uniformly analytic sections by Proposition 1.6. The desired result therefore follows from Proposition 1.2. $\square$

Let $\mathcal{A}$ denote the countable set of approximations. Given $B \subseteq \kappa^{\mathbb{N}} \times X$ and $s \in \kappa^{<\mathbb{N}}$, we say that a is B -terminal on $\mathcal{N}_s$ if a is B_σ -terminal for every $\sigma \in \mathcal{N}_s$, and set $A_s(a, B) = A(a, B) \cap (\mathcal{N}_s \times X)$. If B has uniformly Borel sections and a is B -terminal on $\mathcal{N}_s$, then Lemmas 4.1 and 4.2, together with Proposition 3.4, yield a set $B_s(a, B) \subseteq \kappa^{\mathbb{N}} \times X$ with uniformly Borel sections such that $A_s(a, B) \subseteq B_s(a, B) \subseteq \mathcal{N}_s \times X$ and $B_s(a, B)$ is G -independent.

We recursively construct an increasing sequence $(S^\alpha)_{\alpha < \kappa^+}$ of subsets of $\kappa^{<\mathbb{N}} \times \mathcal{A}$. Set $S^0 = \emptyset$ and $S^\delta = \bigcup_{\alpha < \delta} S^\alpha$ for all limit ordinals

$\delta < \kappa^+$. Given $\alpha < \kappa^+$ and $(S^\beta)_{\beta \leq \alpha}$, define $\rho^\alpha: S^\alpha \rightarrow \alpha$ by $\rho^\alpha(s, a) = \min\{\beta < \alpha \mid (s, a) \in S^{\beta+1}\}$, $B^\alpha = \sim \nabla_{s \in \kappa^{<\mathbb{N}}} \bigcup_{(s,a) \in S^\alpha} B_s(a, B^{\rho^\alpha(s,a)})$, and $S^{\alpha+1} = S^\alpha \cup \{(s, a) \in \sim S^\alpha \mid a \text{ is } B^\alpha\text{-terminal on } \mathcal{N}_s\}$, and fix $B_s(a, B^\alpha)$ as above for all $(s, a) \in S^{\alpha+1} \setminus S^\alpha$. As $|\kappa^{<\mathbb{N}} \times \mathcal{A}| = \kappa$, there is a least $\lambda < \kappa^+$ with $S^\lambda = S^{\lambda+1}$. Set $B = B^\lambda$, $S = S^\lambda$, and $\rho = \rho^{\lambda+1}$.

Lemma 4.3. *Suppose that $(s, a) \notin S$. Then there are $\tau \in \mathcal{N}_s$, a one-step extension b of a , and a τ -configuration γ compatible with b and B_τ such that $(\tau \upharpoonright n^b, b) \notin S$.*

Proof. By the choice of λ , there is $\tau \in \mathcal{N}_s$ for which a is not B_τ -terminal, so there are a one-step extension b of a and a τ -configuration γ compatible with b and B_τ . Set $r = \tau \upharpoonright n^b$. Suppose, towards a contradiction, that $(r, b) \in S$, and set $\beta = \rho(r, b)$. Then b is B^β -terminal on $\mathcal{N}_r$, so $A_r(b, B^\beta) \subseteq B_r(b, B^\beta)$. As $B \subseteq B^\beta$, the configuration γ is compatible with B_τ^β , hence $(\tau, (\phi_\tau \circ \phi^\gamma)(s_{n^b})) \in A_r(b, B^\beta) \subseteq B_r(b, B^\beta)$. The same pair belongs to B , since γ is compatible with B_τ , which contradicts the definition of B . $\square$

Let a_0 denote the unique approximation for which $n^{a_0} = 0$.

Lemma 4.4. *If $(\emptyset, a_0) \notin S$, then there are $\sigma \in \kappa^{\mathbb{N}}$ and a continuous homomorphism from $\mathbb{G}_0$ to G_σ .*

Proof. Recursively construct $s_n \in \kappa^n$ and approximations a_n with $n^{a_n} = n$ and $(s_n, a_n) \notin S$, beginning with $s_0 = \emptyset$ and a_0 . Given s_n and a_n , apply Lemma 4.3 to obtain $\tau_n \in \mathcal{N}_{s_n}$, a one-step extension a_{n+1} of a_n , and a τ_n -configuration γ_n compatible with a_{n+1} and B_{τ_n} , and set $s_{n+1} = \tau_n \upharpoonright n+1$. The same lemma ensures that $(s_{n+1}, a_{n+1}) \notin S$.

Set $\sigma = \bigcup_{n \in \mathbb{N}} s_n$. Then $\tau_n \rightarrow \sigma$. Define $\phi': 2^{\mathbb{N}} \rightarrow \mathbb{N}^{\mathbb{N}}$ and $\psi'_k: 2^{\mathbb{N}} \rightarrow \mathbb{N}^{\mathbb{N}}$ by $\phi'(c) = \bigcup_{n \in \mathbb{N}} \phi^{a_n}(c \upharpoonright n)$ and $\psi'_k(c) = \bigcup_{n > k} \psi_k^{a_n}(c \upharpoonright n - (k+1))$. For all $c \in 2^{\mathbb{N}}$, the sequence $(\tau_n, \phi^{\gamma_n}(c \upharpoonright n+1))$ converges to $(\sigma, \phi'(c))$, so the closedness of C gives $(\sigma, \phi'(c)) \in C$. Similarly, for all $k \in \mathbb{N}$ and $c \in 2^{\mathbb{N}}$, the closedness of D gives $(\sigma, \psi'_k(c)) \in D$.

Define $\pi: 2^{\mathbb{N}} \rightarrow X$ by $\pi(c) = (\text{proj}_X \circ \phi_\sigma)(\phi'(c))$. Passing to the limit in the configuration identities yields $(\text{proj}_{X \times X} \circ \psi_\sigma)(\psi'_k(c)) = (\pi(s_k \cap (i) \cap c))_{i < 2}$ for all $k \in \mathbb{N}$ and $c \in 2^{\mathbb{N}}$. Thus π is a continuous homomorphism from $\mathbb{G}_0$ to G_σ . $\square$

If $(\emptyset, a_0) \notin S$, Lemma 4.4 yields (2). Suppose instead that $(\emptyset, a_0) \in S$, and set $\alpha = \rho(\emptyset, a_0)$. As $A(a_0, B) = B \cap \bigcup_{i < 2} p_i(G)$, $B \subseteq B^\alpha$, and $A(a_0, B^\alpha) \subseteq B_\emptyset(a_0, B^\alpha)$, the definition of B gives $A(a_0, B) = \emptyset$. Thus B is G -independent. For all $a \in \mathcal{A}$ and $n \in \mathbb{N}$, define $S_{a,n} = \{s \in \kappa^n \mid (s, a) \in S\}$ and $B_{a,n} = \nabla_{s \in S_{a,n}} B_s(a, B^{\rho(s,a)})$. Every $B_{a,n}$ is G -independent, since at a fixed parameter at most one member of $S_{a,n}$

contributes. Moreover, $\kappa^{\mathbb{N}} \times X = B \cup \bigcup_{a \in \mathcal{A}, n \in \mathbb{N}} B_{a,n}$. Disjointifying this countable uniformly Borel cover by G -independent sets yields (1).

To see that (1) and (2) are mutually exclusive, note that if both hold then there is a Borel $\mathbb{N}$ -coloring of $\mathbb{G}_0$. $\square$

5. STRATIFICATIONS AND CATEGORY TRANSFORMS

Given a set $B \subseteq \kappa^{\mathbb{N}} \times X$ with uniformly Borel sections and $t \in \kappa^{<\mathbb{N}}$, define $B^{*t} = \{x \in X \mid B^x \text{ is comeager in } \mathcal{N}_t\}$.

Proposition 5.1. *Suppose that κ is an aleph, $t \in \kappa^{<\mathbb{N}}$, X is a topological space, $B \subseteq \kappa^{\mathbb{N}} \times X$ and $B_s \subseteq \kappa^{\mathbb{N}} \times X$ have uniformly Borel sections for all $s \in \kappa^{<\mathbb{N}}$, and $U \subseteq X$. Then:*

- (1) $(\kappa^{\mathbb{N}} \times U)^{*t} = U$.
- (2) $\sim B^{*t} = \sim \bigcup_{u \sqsupseteq t} B^{*u}$.
- (3) $(\bigtriangledown_{s \in \kappa^{<\mathbb{N}}} B_s)^{*t} = \bigcap_{u \sqsupseteq t} \bigcup_{v \sqsupseteq u} \bigcup_{n \leq |v|} B_{v|n}^{*v}$.

Proof. As (1) and (2) are straightforward, we need only establish (3). Towards this end, note that if $x \in (\bigtriangledown_{s \in \kappa^{<\mathbb{N}}} B_s)^{*t}$, then $(\bigtriangledown_{s \in \kappa^{<\mathbb{N}}} B_s)^x$ is comeager in $\mathcal{N}_t$, so $\forall u \sqsupseteq t \exists n \in \mathbb{N} \bigcup_{s \in \kappa^n} \mathcal{N}_s \cap B_s^x$ is non-meager in $\mathcal{N}_u$, thus $\forall u \sqsupseteq t \exists s \in \kappa^{<\mathbb{N}} \mathcal{N}_s \cap B_s^x$ is non-meager in $\mathcal{N}_u$, and it follows that $\forall u \sqsupseteq t \exists v \sqsupseteq u \exists n \leq |v| B_{v|n}^x$ is comeager in $\mathcal{N}_v$, hence $x \in \bigcap_{u \sqsupseteq t} \bigcup_{v \sqsupseteq u} \bigcup_{n \leq |v|} B_{v|n}^{*v}$. And if $x \notin (\bigtriangledown_{s \in \kappa^{<\mathbb{N}}} B_s)^{*t}$, then $(\bigtriangledown_{s \in \kappa^{<\mathbb{N}}} B_s)^x$ is not comeager in $\mathcal{N}_t$, so there exists $u \sqsupseteq t$ such that $(\bigtriangledown_{s \in \kappa^{<\mathbb{N}}} B_s)^x$ is meager in $\mathcal{N}_u$, thus $\forall v \sqsupseteq u \forall n \leq |v| B_{v|n}^x$ is meager in $\mathcal{N}_v$, hence $x \notin \bigcup_{v \sqsupseteq u} \bigcup_{n \leq |v|} B_{v|n}^{*v}$. $\square$

A subset of a topological space is κ^+ -Borel if it belongs to the smallest family containing the open sets and closed under complements and unions of cardinality at most κ .

Proposition 5.2. *Suppose that κ is an aleph, $t \in \kappa^{<\mathbb{N}}$, X is a topological space, and $B \subseteq \kappa^{\mathbb{N}} \times X$ has uniformly Borel sections. Then B^{*t} is κ^+ -Borel.*

Proof. By the obvious induction using Proposition 5.1. $\square$

A set $A \subseteq X$ is κ -Souslin if there exist a closed set $C \subseteq \kappa^{\mathbb{N}}$ and a continuous function $\phi: C \rightarrow X$ with the property that $A = \phi(C)$. We say that a set $A' \subseteq \kappa^{\mathbb{N}} \times X$ is a *stratification* of a κ -Souslin set A if there are witnesses $C \subseteq \kappa^{\mathbb{N}}$ and $\phi: C \rightarrow X$ to the κ -Souslinity of A with the property that $A' = \phi'(C')$, where $C' = \{(\sigma, b) \in \kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}} \mid \sigma \circ b \in C\}$ and $\phi': C' \rightarrow \kappa^{\mathbb{N}} \times X$ is given by $\phi'(\sigma, b) = (\sigma, \phi(\sigma \circ b))$. For each $\sigma \in \kappa^{\mathbb{N}}$, we view A'_σ as the *analytic trace* of A on the set $\sigma(\mathbb{N})$.

Proposition 5.3. *Suppose that κ is an aleph, X is a Hausdorff space, $A \subseteq X$ is κ -Souslin, $A' \subseteq \kappa^{\mathbb{N}} \times X$ is a stratification of A , and $x \in A$. Then $\forall^* \sigma \in \kappa^{\mathbb{N}}$ $x \in A'_\sigma$.*

Proof. Fix a closed set $C \subseteq \kappa^{\mathbb{N}}$ and a continuous function $\phi: C \rightarrow X$ with the property that $A = \phi(C)$ and $A' = \phi'(C')$, as well as $\tau \in C$ for which $\phi(\tau) = x$. Then the set $H = \bigcap_{n \in \mathbb{N}} \bigcup_{s \in \kappa^{< \mathbb{N}}} \mathcal{N}_{s \smallfrown (\tau(n))}$ is a dense G_δ subset of $\kappa^{\mathbb{N}}$ and $\forall \sigma \in H$ $x \in A'_\sigma$. $\square$

Proposition 5.4. *Suppose that κ is an aleph, $t \in \kappa^{< \mathbb{N}}$, X is a Hausdorff space, G is a κ -Souslin digraph on X , G' is a stratification of G , $B \subseteq \kappa^{\mathbb{N}} \times X$ has uniformly Borel sections, and B_σ is G'_σ -independent for every σ . Then B^{*t} is G -independent.*

Proof. Otherwise fix $x, y \in B^{*t}$ with $x G y$ and appeal to Proposition 5.3 to obtain $\sigma \in \mathcal{N}_t$ with $x, y \in B_\sigma$ and $x G'_\sigma y$, a contradiction. $\square$

Theorem 5.5 (Kanovei). *Suppose that κ is an infinite cardinal, X is a Hausdorff space, and G is a κ -Souslin digraph on X . Then at least one of the following holds:*

- (1) *There is a κ^+ -Borel coloring $c: X \rightarrow \kappa$ of G .*
- (2) *There is a continuous homomorphism from $\mathbb{G}_0$ to G .*

Proof. Fix a stratification G' of G . As every homomorphism from $\mathbb{G}_0$ to a vertical section of G' is a homomorphism from $\mathbb{G}_0$ to G , Theorem 1 allows us to assume there is a uniformly Borel function $c': \kappa^{\mathbb{N}} \times X \rightarrow \mathbb{N}$ such that $\forall \sigma \in \kappa^{\mathbb{N}}$ c'_σ is a coloring of G'_σ . For all $n \in \mathbb{N}$, set $B_n = (c')^{-1}(\{n\})$. For all $t \in \kappa^{< \mathbb{N}}$, Proposition 5.2 ensures that B_n^{*t} is κ^+ -Borel and Proposition 5.4 ensures that it is G -independent. It only remains to note that if $x \in X$, then $\kappa^{\mathbb{N}} = \bigcup_{n \in \mathbb{N}} B_n^x$, so there exists $n \in \mathbb{N}$ for which B_n^x is non-meager. Proposition 2.1 and localization therefore yield $t \in \kappa^{< \mathbb{N}}$ for which B_n^x is comeager in $\mathcal{N}_t$, thus $x \in B_n^{*t}$. $\square$

Acknowledgements. ChatGPT provided substantial assistance in the discovery and development of the results.

REFERENCES

- [dRM21] N. de Rancourt and B. D. Miller, *The Feldman–Moore, Glimm–Effros, and Lusin–Novikov theorems over quotients*, 2021.
- [Kan97] Vladimir Kanovei, *Two dichotomy theorems on colourability of non-analytic graphs*, *Fundamenta Mathematicae* **154** (1997), no. 2, 183–201.
- [Kec95] Alexander S. Kechris, *Classical descriptive set theory*, Graduate Texts in Mathematics, vol. 156, Springer-Verlag, New York, 1995.